CARLOS ROMERO MADRID

COMPUTATIONAL PHYSICIST | SCIENTIFIC SOFTWARE ENGINEER | CREATIVE COMPUTING

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✦ LIVE INTERACTIVE WEB DOSSIER & DIGITAL EXPERIENCES

Explore online: Interactive European Trajectory Map · Real-Time MN5 Telemetry Audio Synthesizer (MNusic) · 3D Bio-Kinetic Sensor Visualizer

INTERACTIVE carrommad.github.io/cool-cv ↗

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01 SCIENTIFIC & ENGINEERING TRAJECTORY

European HPC & Applied Research

Barcelona Supercomputing Center (BSC-CNS) — MAY 2025 – PRESENT
HPC Scientific Software Engineer · Performance Team — Barcelona, Spain
Contributing to ocean–atmosphere coupling in the **DestinationE** and **ClimateDT** digital twin projects across European tier-0 infrastructure. Developing new capabilities in the **ECLand** climate model alongside the Climate Variability and Change team. Deploying, configuring, and orchestrating scalable simulation workflows on **MareNostrum 5 (MN5)**.

Barcelona Supercomputing Center (BSC-CNS) — MAY 2023 – APR 2025
Research Engineer · Fusion Team (Advanced Computing Hub) — Barcelona, Spain
Optimized parallelization of nuclear fusion simulation codes using **OpenMP, MPI, PETSc, oneAPI, OpenACC** on European Tier-0 supercomputers. Implemented 3D electromagnetic modeling for superconducting fusion magnets in the **MAGNET** module as part of the **ALYA** multi-physics system; led code reviews and technical documentation.

National Renewable Energy Centre (CENER) — MAY 2022 – SEP 2022
Data Scientist · Analysis and Design — Pamplona, Spain
Engineered Python data pipelines for real-time characterization and control of offshore wind plants. Built automated predictive maintenance classifiers with **Pandas & Scikit-Learn**, translating multi-terabyte sensor streams into operational maintenance decisions.

University of Seville — SEP 2020 – APR 2022
Research Scientist · Plasma & Fusion Technology — Seville, Spain
Conducted theoretical and computational plasma physics on HPC clusters using gyrokinetic theory and MHD models. Built Python diagnostics for 3D charged-particle tracking and Fortran magnetic equilibria solvers. Taught undergraduate Quantum Physics.

Fraunhofer Institute for High Frequency Physics (FHR) — AUG 2018 – APR 2019
Research Scientist · Cognitive Radar — Bonn / Wachtberg, Germany
Formulated simulated annealing optimization algorithms in MATLAB for non-planar antenna array geometries, boosting directivity and angular accuracy in next-generation automotive radar systems.

02 ACADEMIC RIGOR

Physics · Math

BSc in Physics & BSc in Mathematics
University of Seville · 2013 – 2018 (Double Degree)
Grade: 8.35 / 10 · 3rd of Class

MSc in Plasma & Fusion Physics
Aix-Marseille University, France · 2019 – 2020
Grade: 15.43 / 20 · 1st of Class (Major)

MSc in Advanced Mathematics
UNED · 2023 – Present (ABD)
Functional Analysis & Complex Systems

03 COMPUTE & HPC STACK

Tier-0 Systems

SUPERCOMPUTERS DEPLOYED

MareNostrum 5 (MN5) | Leonardo (CINECA) | LUMI (CSC)

PARALLEL & SCIENTIFIC LANGUAGES

Fortran | C / C++ | Python | MPI | OpenMP | CUDA | OpenACC | PETSc | Slurm | oneAPI | Bash/Shell | Git

04 CREATIVE COMPUTING

Sound · Motion · Tangible

MNusic Synthesizer — Web Audio API
Real-time sonification converting live MareNostrum 5 & LUMI cluster telemetry into interactive harmonic soundscapes.

Pipa-Music-Ball — ESP32 · 6-DoF
Bio-kinetic sensor sphere turning feline movement into microtonal scales. Interactive 3D visualizer available in web dossier.

Expo-Curator — Open Source
Gallery curation engine for automated artwork classification, catalog layout, and exhibition wall staging.

SPOKEN LANGUAGES: Spanish (Native) · English (C1) · French (C1) · Italian & German (Inter)

EUROPEAN RESEARCH JOURNEY: **Seville → Bonn → Marseille → Pamplona → Barcelona**

INTERACTIVE DOSSIER: **carrommad.github.io/cool-cv**